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320413 (20)

BE (4th Semester)

Examination, April- May, 2010

Branch : Civil

SURVEYING – II

Time allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Answer questions from all units. Part–A of all units is compulsory (2 marks). Answer any two questions from Part–B (2 × 7 = 14).

[(2 + 14)5 = 80]

Unit-I

Part–A

Q. 1. (a) What is the angle value for a well conditioned Isoscles triangle. 2

Part-B

(b) (1) Derive the condition for satellite station "reduction to centre" when the satellite station is to the inside of main triangulation stations ? 7

(2) From an eccentric station S, 12.25 m to the west of the main station, B, the following angles were measured

$\angle BSC = 76°25'32''$

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∠CSA = 54°32'20". The stations S and C are to the opposite sides of line AB. If AB & BC are 5286.5 m and 4932.2 m find the correct angle ABC ? 7

(3) Explain the terms

1. Reconnaissance

2. Base line selection 7

Unit-II

Part–A

Q. 2. (a) Define an "accidental error" ? 2

Part-B

(b) (1) Find MPV of A by Normal Equations.

A = 30° 28' 40"

3A = 91° 25' 55"

4A = 121° 54' 30" 7

(2) Solve the MPV of angles by method of Correllates ?

AOB = 85° 42' 28.75" wt 3

BOC = 102° 15' 43.26" wt 2

COD = 94° 38' 27.22" wt 4

DOA = 79° 23' 24.77" wt 2 7

(3) A surveyor carried out levelling operations of a closed circuit ABCDA

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starting from A and made the following observations.

B was 8.164 m above A, wt 2.

C was 6.284 m above B, wt 2

D was 5.626 m above C, wt 3

D was 19.964 m above A, wt 3

Determine the probable heights of BC & D above A ? 7

Unit-III

Part–A

Q. 3. (a) List the normal values of additive constant and multiplicative constant in tacheometry ? 2

Part-B

(b) (1) Derive the tacheometric constants in a theodolite ? 7

(2) Derive the formulae for horizontal distance and vertical distance in a tacheometric surveying with staff vertical, sight inclined ? 7

(3) A tacheometer is set up at an intermediate point on a traverse PQ 7

Staff St^{u}	Vertical Angle	Staff Intercept	Hair Readings
P	+8° 36'	2.350	2.105
Q	+6° 6'	2.055	1.895

Unit-IV

Part–A

Q. 4. (a) Define "Principal Point" of a photograph ? 2

Part-B

(b) (1) Derine focal length of a camera lens ? 7

(2) Define : 1. Oblique Photograph

2. Nadir Point

3. Iso centre 7

(3) Derive various scales of a photograph ? 7

Unit-V

Part–A

Q. 5. (a) List the different Tide Gauges ? 2

Part-B

(b) (1) Derive the three point problem in hydrographic survey by Analytical Solution ? 7

(2) Explain the 2 mechanical methods of solution of 3 point problem in hydrographic surveying ? 7

(3) Explain the location of soundings by :

1. Rangle and One Angle from shore
2. Range and Two Angles from shore ? 7

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